

AI Business Analyst Process Management

Assignment Submission — DeepThought Fellows Programme

PharmaTech Ingredients Pvt. Ltd. — Operations Diagnostic

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PART A | Diagnosis: What Is Actually Going On

For each situation I identify the stated problem, the real underlying dynamic, and cite specific evidence from the scenario. Diagnosis comes before prescription.

A1 The Alert Nobody Responds To

STATED PROBLEM

The daily automated email goes out every morning. The CEO sees red items. The same items stay red for days. Anil and Karthik say they send the report every day and cannot understand why no one acts on it. The CEO says he does not have visibility.

REAL DYNAMIC

The alert is a broadcast with no addressee, no context, and no decision prompt. It reports status to the people who already know the status, and asks nothing of anyone specifically. It is a dashboard, not a trigger. Three structural failures compound each other.

1. No individual accountability. The email lists departments — Purchase RED, Utilities RED — not people. When a problem belongs to a department it belongs to no one. No one reads 'Purchase RED' and feels personally responsible for Monday morning.

2. No escalation pathway or decision prompt. The email tells the CEO what is red but gives him no handle — no name to call, no action pre-suggested, no history of whether this was red last week. Seeing a red item with no escalation path is like seeing a fire alarm with no exit signs. The natural response is to wait.

3. No trend or memory. The system resets daily. There is no signal that Purchase has been red for three consecutive weeks. Without trend data, a single-day red feels like a normal operational blip, not a systemic failure. The CEO cannot distinguish chronic dysfunction from a one-day delay.

Beneath these structural problems is a behavioural one: because the email never named a person, and because the CEO's visible response to red items has been to wait (no escalation call, no visible consequence to HODs), the HODs have learned that red on the email carries no cost. The system trained them to ignore it.

Evidence from scenario: The alert email shows department names (Purchase, Warehouse, QC Lab), not individual names. The senior advisor's direct observation: 'If I see Warehouse delayed, who feels the pressure? Nobody. If I see Dinesh, 3 tasks, 1 red, now Dinesh comes prepared tomorrow.' The email also shows no days-overdue trend, no escalation history, and no 'changed since yesterday' field. The CEO's own words confirm the system fails him: the same items stay red for days; nobody responds or escalates; he wants to know what is happening on the floor, not what happened yesterday.

A2 The Blank Root Cause Field

STATED PROBLEM

The deviation tracker has a root cause field. 70% of entries are blank or filled with generic placeholders such as 'under investigation' or 'process deviation'. Anil and Karthik say people are not disciplined enough to fill it.

REAL DYNAMIC

This is not a discipline problem. It is a fear-of-blame problem that has been reinforced by at least one concrete, visible incident of public humiliation. The supervisors are making a rational choice: the personal cost of writing an honest root cause is higher than the cost of leaving the field blank.

Supervisor 2 described a specific event — the maintenance head was called out in front of everyone when an honest root cause was documented. That single incident sent a message to every other supervisor: transparency gets you punished. The blank field is not laziness; it is a learned defensive behaviour.

Supervisor 1 made the calculation explicit: if he writes what actually went wrong, the next review meeting turns into 50 questions and a blame game. So he leaves it blank and handles it quietly. This is a person doing their job — protecting themselves — in an environment where the system incentivises concealment over honesty.

Why did Anil and Karthik miss it? They diagnosed from the top down. They observed the output (blank fields) and interpreted it through the lens of a process manager: non-compliance = indiscipline. They never sat with a supervisor and asked why. Their role drift — spending time on follow-ups and fixing printers — left no bandwidth for diagnostic conversations. The existing power dynamic also makes it unlikely a supervisor would voluntarily tell a 25-year-old analyst that he is afraid of being blamed in front of his colleagues.

Evidence from scenario: Supervisor 1: 'If I write what actually went wrong, tomorrow in the review meeting they will ask me 50 questions... it becomes a blame game. So I leave it blank and handle it quietly.' Supervisor 2: 'Last time I wrote the real root cause... the maintenance head got called out in front of everyone. After that, nobody writes root cause honestly.' The deviation table sample shows 5 of 7 root cause fields blank or generic. Anil and Karthik's stated diagnosis (not disciplined enough) directly contradicts what supervisors said when spoken to separately.

A3 The Meeting That Eats the Day

STATED PROBLEM

The daily status meeting is scheduled for 30 minutes. It runs 60–90 minutes. After the meeting, everyone's day is derailed by unplanned work that surfaced during it. Anil and Karthik say technical issues come up that have to be discussed.

REAL DYNAMIC

This is not a meeting problem. It is a planning vacuum that has turned the meeting into the only place where operational reality becomes visible. The meeting functions as a de facto planning session, deviation review, technical forum, and status update — all at once — because those functions do not exist anywhere else in the operating rhythm.

The core issue: nobody in the company has a daily micro-plan. The senior advisor confirmed this directly — there is a monthly plan and a weekly plan, but no daily plan at any level, not for Anil, Karthik, or the HODs. When there is no plan for the day, every deviation from the default is unplanned. The organisation cannot distinguish a routine task from a project activity from a genuine emergency — because they are all equally unscheduled.

The meeting then becomes the discovery mechanism for the day's reality. Because 12 HODs each need 5–10 minutes to surface their issues (two hours of content compressed into 30 minutes), the meeting structurally cannot finish on time. Technical issues surface because there is no separate escalation channel — an OOS lab result has nowhere else to go, so it takes up group time in the only shared forum.

The downstream consequence — derailed afternoons — is the natural result. When the meeting surfaces unplanned work with no pre-assigned owner and no triage protocol, the default response is everyone involved figures it out now. This generates reactive, scattered afternoons across all 12 departments every single day.

Evidence from scenario: Senior advisor: '12 people x 10 minutes = 120 minutes. Why is this meeting 30 minutes?' — silence. Confirmed: monthly plan exists, weekly plan exists, no daily plan. 'If there is no plan for today, everything is unplanned.' Advisor identified three activity types (routine, planned, unplanned) that nobody was categorising or tracking. Technical issues such as OOS results discussed in the main status forum — no separate escalation channel exists. Anil's own words: most of his time goes to follow-ups; by the time he is done following up, there is no bandwidth to think or build.

PART B | Intervention Design: What Would You Actually Do

For each situation: the first conversation, two-week human and process changes, the tool that reinforces the change, and how to earn buy-in from a 15-year floor veteran.

B1 Intervention — The Alert System

First Conversation

Not with the CEO. With one HOD who has a red item that week — say, the Purchase head whose four POs have been pending for three days. I sit next to him informally (not in a meeting) and ask one question: *When you see that email come in the morning, what do you actually do with it?* I let him answer fully. I am listening for whether he feels the email is addressed to him, whether he knows what is expected of him when he sees red, and whether he has ever received a follow-up from anyone about a red item.

My hypothesis is he will say he sees it, knows it is red, and is working on it. That tells me the signal is reaching him — the problem is the absence of a response mechanism. I then ask: *If I marked a task specifically to you — your name, one action, a deadline — would that feel different?* This plants the idea without announcing a system change.

First Two Weeks (No Tools Yet)

- **Days 1–3:** Same informal conversation with 3–4 more HODs. I want to know who reads the email seriously and who treats it as noise. I shadow one daily status meeting without speaking — watching who reports what, how the CEO or COO responds to red items, and whether any red item from last week reappears.
- **Days 4–5:** Short conversation with the COO (Mr. B). I frame it as a question: *When you see a red item in the morning email, what is your next step?* If he says he does not always follow up, I now understand the accountability gap is structural, not just behavioural.
- **Week 2:** Without building any new tool, I manually pilot a change: for two departments, I add individual names next to tasks in the existing tracker. I observe whether HODs come to the next meeting more prepared on those items. This is a behavioural test before I build anything.

Tool Design

What the HOD sees: The daily email changes from a department-level table to a personal one. Each HOD receives their own version: only their items, with their name at the top. Subject line: *[Name] — 2 items need your attention today*. Three columns: Task | Days Overdue | Your Next Step (pre-filled suggestion, e.g. Approve PO 1042 today). One sentence at the bottom: Reply to this email with a status update by 3pm.

CEO/COO version: They continue to receive the aggregate view, now with individual names against each item, a streak column showing how many consecutive days an item has been red, and a one-line escalation flag for anything overdue five or more days.

Workflow: Google Sheets tracker (already exists) → Power Automate filters rows by HOD name →

sends personalised email via Outlook. Replies are CC'd to the COO's inbox, not routed to a system — because the COO already does daily follow-up.

Technology: Power Automate (already in use), existing Google Sheets, Outlook. No new tool introduced. The only change is routing logic and personalisation of output.

Buy-In from a 15-Year Veteran

I would not walk into his cabin and explain the new system. I would find him after a morning meeting and say: *Sir, I noticed last week your Purchase items kept showing up in the group email but you were already working on them. That must be frustrating — getting flagged for something you are already handling. I am trying to fix that. Would it help if the email only showed your items, so it is actually useful to you in the morning rather than noise?*

This acknowledges his competence, frames the change as serving him rather than monitoring him, and asks his opinion rather than informing him of a decision. If he says yes — I have buy-in and a champion. If he says he does not read the email at all, I have learned something more important.

B2 Intervention — The Deviation Tracker

First Conversation

With a supervisor — not a HOD, not Anil or Karthik — specifically one who has a recent deviation entry with a blank root cause field. I speak privately, one-on-one, away from the floor: *I saw this deviation from last week. I am not asking who caused it. I am trying to understand what actually happened — because if I do not understand it, I cannot help prevent it from happening again. What actually went wrong?*

I listen. I do not write anything down in front of them. I do not ask who was responsible. I am only confirming my hypothesis (that they know the answer and chose not to write it) and building the trust needed before I change anything. If they tell me the real root cause, I say: *That is exactly the kind of thing I want the system to capture, without making it a blame session for anyone.*

First Two Weeks (No Tools Yet)

- **Conversation with CEO/COO:** I flag that blank root cause fields are not a discipline problem but a psychological safety problem. I specifically mention the maintenance head incident. I recommend one explicit norm change: root cause review becomes a learning discussion, not an accountability hearing. The HOD is asked 'What do we need to fix?' not 'Why did your team fail?'
- **Norm signal in the next review meeting:** If the CEO or COO publicly thanks the first person who writes an honest root cause — 'This is useful, now we can actually fix it' — that one moment does more than any training session. I ask Anil or Karthik to watch for and encourage this.
- **Separate technical triage channel:** Before building a new form, I create a norm: deviation reviews happen in a separate 15-minute call between QA, Production, and Maintenance — not in the main daily status meeting. This removes the public-humiliation context that caused the original incident.

Tool Design

What the supervisor sees: A short WhatsApp message from an automated number within two hours of a deviation being logged: *Deviation logged: [Batch 247 OOS]. To close this by tomorrow, please answer: 1) What do you think caused this? (one sentence is fine) 2) What should we change to prevent it? Your response will only be reviewed in the technical meeting — not in the daily status call.*

This is deliberately low-friction (one to two sentences on WhatsApp), explicitly private (not the daily meeting), and framed as future-focused (what should we change) rather than backward-looking (who caused this).

Workflow: Deviation logged in existing tracker → Power Automate triggers WhatsApp message (via WhatsApp Business API or Twilio) → supervisor responds via linked Google Form → response populates root cause field in tracker → aggregated in a separate Learning Log visible to QA head and COO only, not in the public daily email.

Technology: WhatsApp (supervisors already use it), Google Forms (zero adoption friction), Power Automate for trigger logic. The Learning Log is a separate tab in the existing Google Sheet — no new tool, no new login.

Buy-In from a 15-Year Veteran

I would find the most respected senior supervisor on the floor — the one others watch. I would not explain the system. I would say: *You have been here long enough to know which problems keep coming back. I am building something that captures that — what actually causes things to go wrong — so we stop solving the same problem three times a year. I want your input on how it should work, because if it does not work for you, it will not work for anyone.*

I then show him a blank version of the WhatsApp prompt and ask him to rewrite it in the language he would actually respond to. Whatever he writes becomes the template. He is now a co-author of the system, not a subject of it.

B3 Intervention — The Daily Meeting

First Conversation

With the COO (Mr. B), before changing anything about the meeting. Not 'the meeting is broken.' Instead: *I have been watching the daily meeting. By the time it ends, everyone's morning is gone and the afternoon is reactive. Can I ask — what do you need from that meeting that you are not currently getting? And what is in it that you could get another way?*

This surfaces his actual requirements. My hypothesis: he wants to know what is blocked and who is handling it — not a 12-person status tour. If he confirms that, I propose a structural change as a question: *What if technical issues went to a separate forum, and the daily call became only exceptions — things that need your decision?*

First Two Weeks (No Tools Yet)

- **Pre-meeting standup (no tool — just a norm):** Each HOD submits three lines before the meeting: what I completed yesterday, what I am doing today, what is blocking me. Format: WhatsApp message to a group, or a card handed to Anil. Purpose: the facilitator reads all 12 before the meeting and compresses it to 20 minutes by only surfacing blockers.
- **Meeting restructure:** Daily status meeting: 30 minutes, exceptions only, no technical issue discussion. Separate technical triage: 15 minutes, Tuesday and Thursday, QA and Production and Maintenance only. Proposed to the COO as a two-week pilot.

- **Activity categorisation:** Anil and Karthik start tagging each task in the tracker as R (Routine), P (Planned), or U (Unplanned) — just two extra characters per row. After one week, the pattern of unplanned work becomes visible. This is the foundation for the tool.

Tool Design

What the HOD sees: A simple daily planning card — Google Form, takes 3 minutes — sent at 8am via WhatsApp: *Good morning. Before the 9am meeting, please log: 1) Top 3 tasks for today. 2) Any blockers or deviations. 3) Is anything from yesterday unfinished?* Dropdown fields for task category (Routine / Planned / Unplanned) and a text field for each item. No more than five fields total.

What Anil sees: A live dashboard (single Google Sheet) that aggregates all 12 HOD inputs by 8:45am. Unplanned items auto-highlighted yellow, blocked items red. Anil walks into the 9am meeting with a prepared agenda — only red and yellow items — instead of discovering the day's problems live in the meeting.

What the CEO sees: The aggregate tab shows, for the first time, what percentage of the company's daily work is unplanned vs planned. Over four weeks, this trend line becomes one of the most diagnostic data points available.

Technology: WhatsApp (existing), Google Forms (zero adoption friction), Google Sheets (existing), Power Automate (existing). No new software. No new logins.

Buy-In from a 15-Year Veteran

I would find the HOD whose team is most often interrupted mid-task when new work emerges from the meeting. I would say: *I have noticed that after the daily meeting, your team gets pulled in three directions. I am trying to reduce that. If we could agree on the day's work before the meeting, would that help you?*

I then show him the three-question WhatsApp format and say: *It is three minutes. If it saves thirty minutes of back-and-forth in the afternoon, is it worth trying for one week?* I do not ask him to believe in the system. I ask him to try it for one week. One week is a low-commitment experiment, not a permanent change.

PART C | Scale Thinking: What Breaks at 8x

The Single Most Dangerous Failure Point

The absence of individual accountability infrastructure.

At 2 projects, the organisation runs on tacit knowledge: the COO knows which HOD is reliable, Anil knows which supervisor actually follows through, the CEO mentally tracks which departments are chronically red. The system works — barely — because a handful of people carry the accountability structure in their heads.

At 8 projects, this collapses. There are too many cross-functional dependencies, too many new clients with unique requirements, and 3–4 new analysts who do not yet have the tacit knowledge that Anil and Karthik accumulated over six months. The new analysts will make the same mistake — diagnosing from the surface, building tools before understanding behaviour — unless the institutional knowledge is codified.

The deviation tracker failure illustrates the stakes precisely. At 2 projects, a blank root cause field is an operational irritant. At 8 projects — with 8 active client batches running simultaneously under regulatory scrutiny — a blank root cause field in one batch can cascade into a failed client audit, a batch rejection, or a regulatory non-conformance across all units. The psychological safety problem that produces blank fields will not self-correct as the organisation scales. It will get worse as more people, more projects, and more review meetings amplify the blame-game dynamic.

The System to Build Now, at 2 Projects

An Individual Accountability and Learning Log — built in the first 3 months.

This is not a new tool. It is a layer added to the existing unified tracker and deviation system. Three components:

1. Named ownership. Every task, every deviation, every planned activity in the tracker has a person's name — not a department name. This is a data hygiene change, not a software change. It can be done in the existing Google Sheet today.

2. Weekly pattern review (the Learning Meeting). A 30-minute weekly meeting separate from the daily status meeting, attended by Anil and Karthik plus the 3–4 HODs with the most unplanned work that week. Agenda: three questions only — What unplanned work came up? What caused it? What can we schedule in advance to prevent it next time? Output: a single row in a Pattern Log: Date | Unplanned Event | Root Cause | Preventive Action | Owner.

3. Analyst onboarding pack. Every week, Anil and Karthik document one thing they learned about the organisation's behaviour — one paragraph in a shared Notion doc. When the new analysts join, this is their first reading material. It is the institutional memory that currently exists only in their heads.

At 8 projects, this system means: new analysts know which HODs need daily prompts vs weekly check-ins; the CEO can see which project has the highest rate of unplanned deviations; and the deviation tracker has real root causes that prevent repeat failures across multiple client batches.

What to Automate, What to Keep Human — and Why

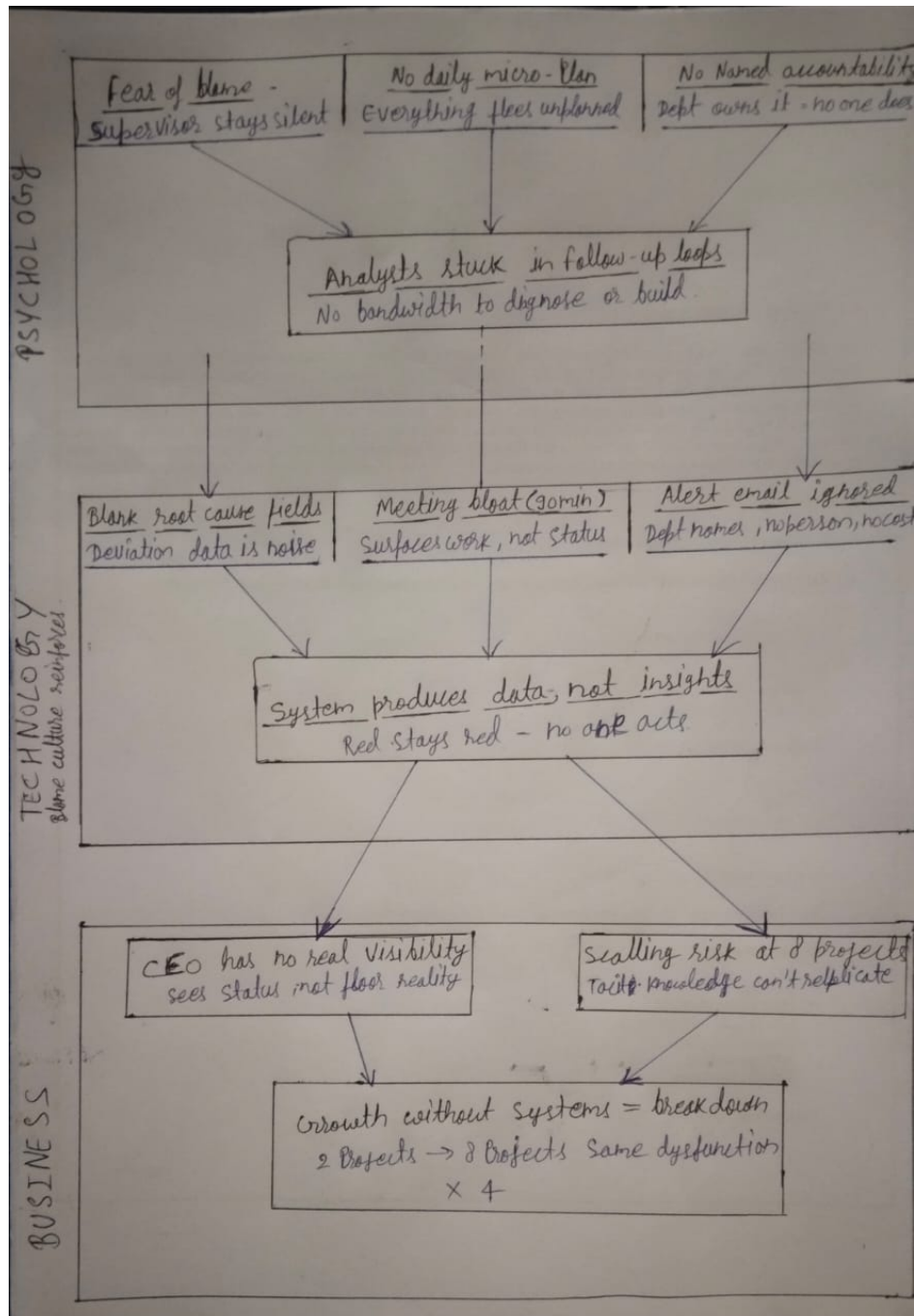
Automate	Keep Human	Reason
Daily personalised alert emails to HODs		Purely mechanical routing. Frees Anil from the follow-up loop.
WhatsApp prompt to supervisor after deviation is logged		Trigger is deterministic (event occurs = message sent). Human sending it adds no value.
Aggregating HOD pre-meeting inputs into a dashboard		Data aggregation is mechanical. The value is in the agenda it produces for Anil.
Trend flagging: Purchase has been red for 5 consecutive days		Pattern recognition across time is where automation earns its place.
	First conversation with any HOD or supervisor about why something is not working	The real information lives in what people will not write. Only a human can create the trust to surface it.
	Root cause review discussions	Psychological safety is a human problem. An automated prompt cannot replace a facilitator who redirects blame toward learning.
	Deciding which unplanned events to escalate to the CEO	Context and judgment are required. The COO knows which red item is a one-day blip and which is a systemic signal.
	Analyst onboarding and knowledge transfer	Tacit knowledge of the organisation — who trusts whom, which HOD needs hand-holding — cannot be automated. It must be documented by humans who experienced it.

The principle behind the line: Automate the mechanical (routing, triggering, aggregating). Keep human the interpretive (diagnosing, facilitating, deciding). The test is: does this task require understanding of human context to do correctly? If yes, keep it human. If no, automate it so the analyst has bandwidth to do the human work.

Note on Part D: The hand-drawn diagram connecting the psychology, technology, and business layers across all three situations is submitted separately as a photograph.

PART D | Hand-Drawn Diagram: The Full Picture

The diagram below was drawn by hand on paper and photographed. It captures the interconnection between all three situations diagnosed in Part A, mapped across the three layers the assignment specifies: psychology, technology, and business.



What the diagram shows

The diagram is organized into three horizontal bands — Psychology, Technology, and Business — with arrows flowing downward to show how each layer produces the next layer's dysfunction.

Psychology layer (top band)

Three root causes sit at the top: fear of blame (supervisors stay silent), no daily micro-plan (everything feels unplanned), and no named accountability (the department owns it, so no one does). All three arrows converge on a single box: analysts stuck in the follow-up loop with no bandwidth to diagnose or build.

Technology layer (middle band)

Each psychology failure produces a technology failure directly below it: fear of blame produces blank root cause fields; the planning vacuum produces meeting bloat (90 minutes of discovering work rather than reporting it); and the absence of named accountability produces an alert email that is ignored because it lists departments, not people. All three converge on: the system produces data, not insight — red stays red, no one acts.

Business layer (bottom band)

The technology failure feeds two business outcomes: the CEO has no real visibility (he sees status, not floor reality) and the organisation carries a scaling risk at 8 projects (tacit knowledge cannot replicate). Both converge on the final box: growth without systems equals breakdown — 2 projects scaling to 8 projects with the same dysfunction, multiplied by four.

The feedback loop

A dashed arrow runs up the left margin from the bottom back to fear of blame at the top, labelled: blame culture reinforces itself. This is the critical systemic insight — the breakdown at scale does not just repeat the original dysfunction; it amplifies it. More projects, more review meetings, more public accountability hearings, more blank fields.

The diagram intentionally does not show three separate problems with three separate solutions. It shows one system with one root dynamic — a culture where transparency carries personal cost — expressing itself across three visible symptoms. The intervention logic follows: fix the psychology first, then the technology reinforces the change, and the business outcome follows.